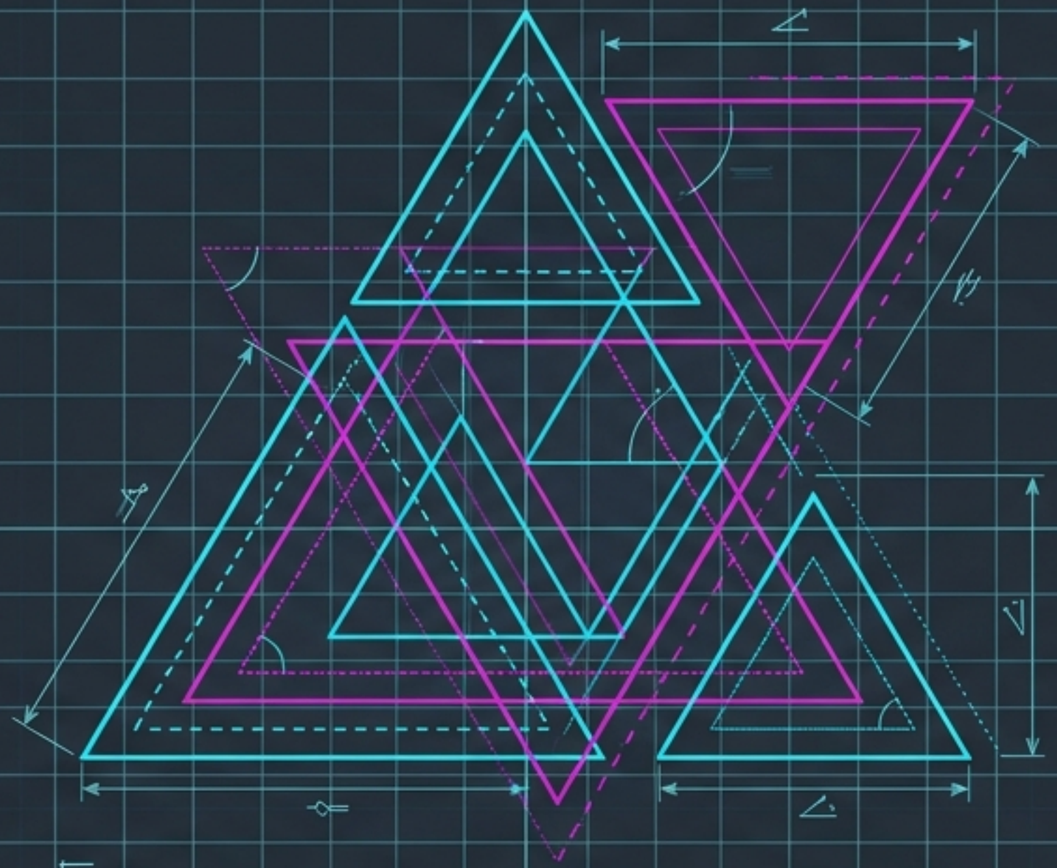
Class X Mathematics Chapter 6: Triangles



Chapter 6: Triangles

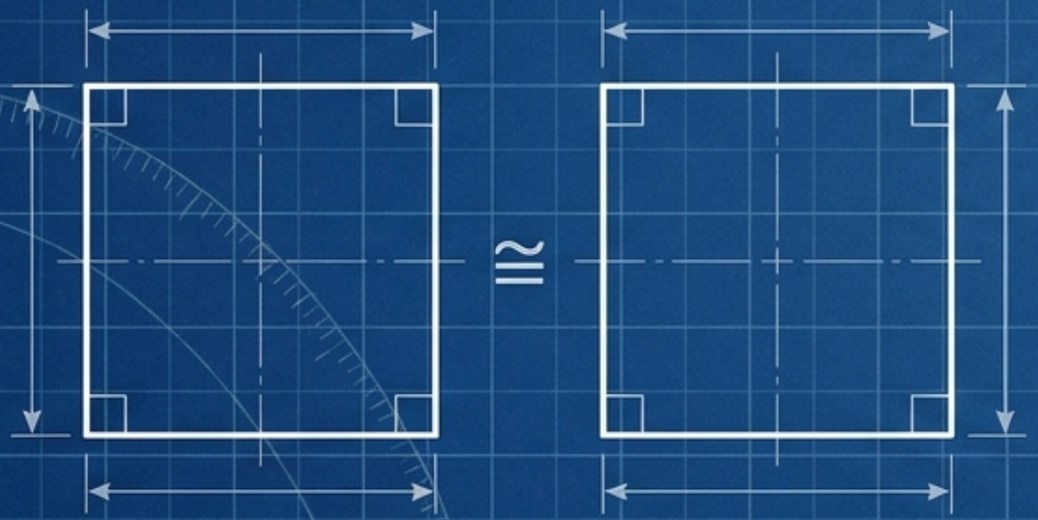
Core Concepts: Similarity, Basic Proportionality, Similarity Criteria, and Pythagoras Theorem.

Gemini Notebook

DEFINING THE RELATIONSHIP BETWEEN SHAPE AND SIZE

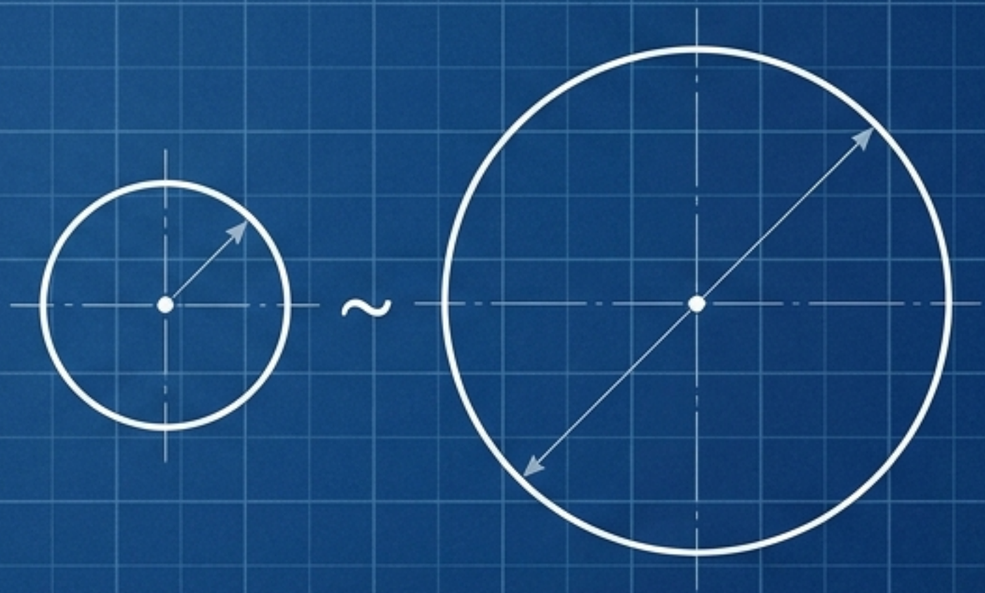
CONGRUENT FIGURES

Same shape AND same size.



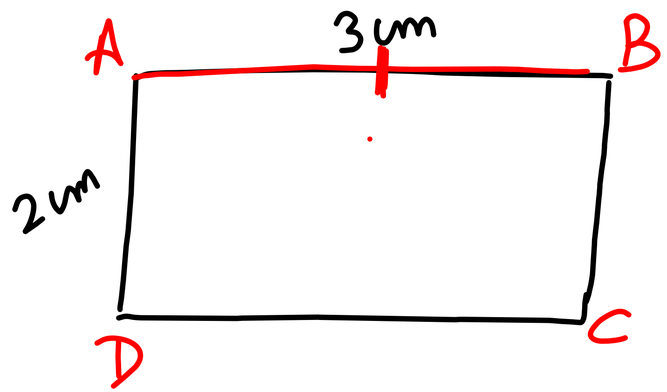
SIMILAR FIGURES

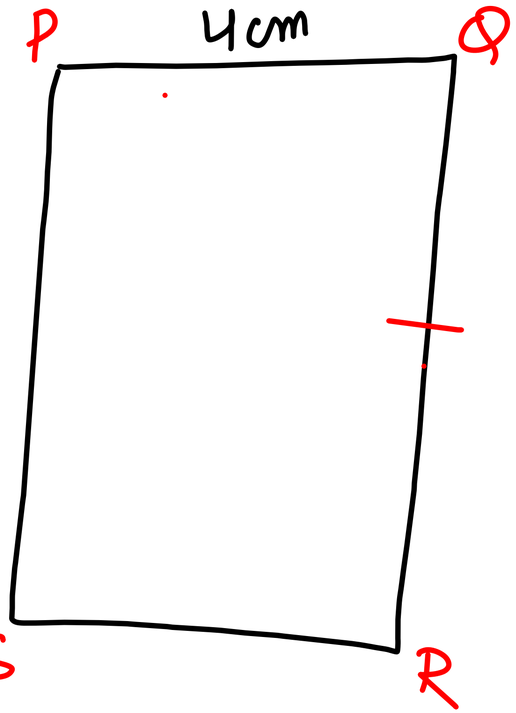
Same shape but NOT necessarily the same size.



All congruent figures are similar, but similar figures need not be congruent.
All circles are similar. All squares are similar.

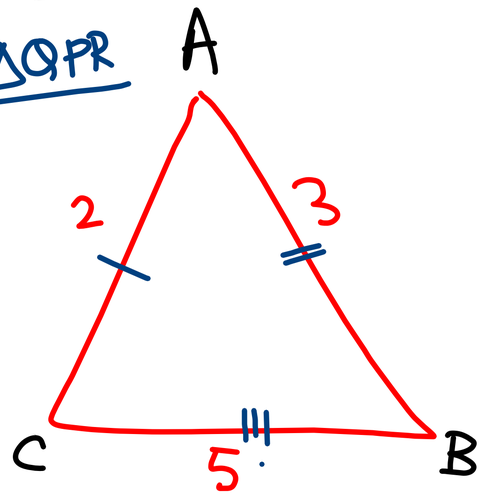
Gemini Notebook

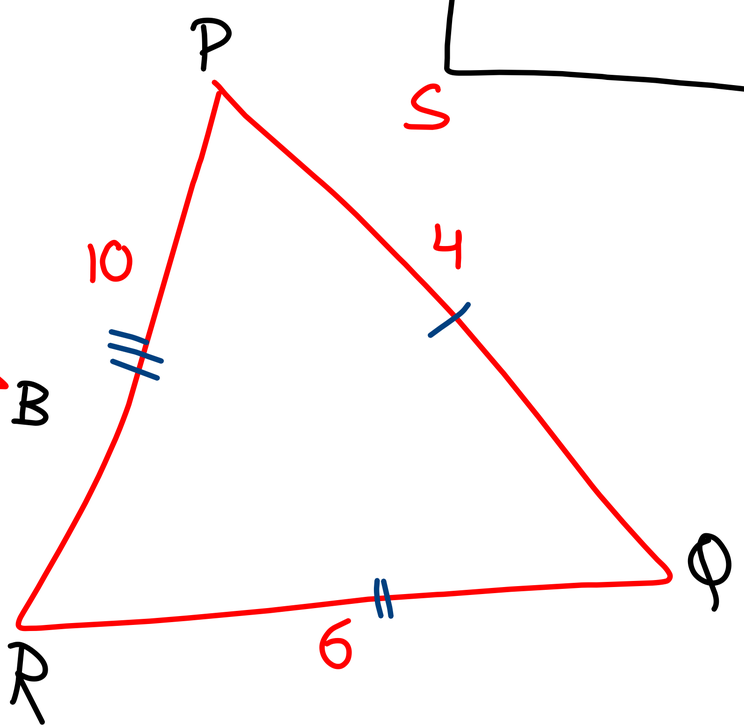




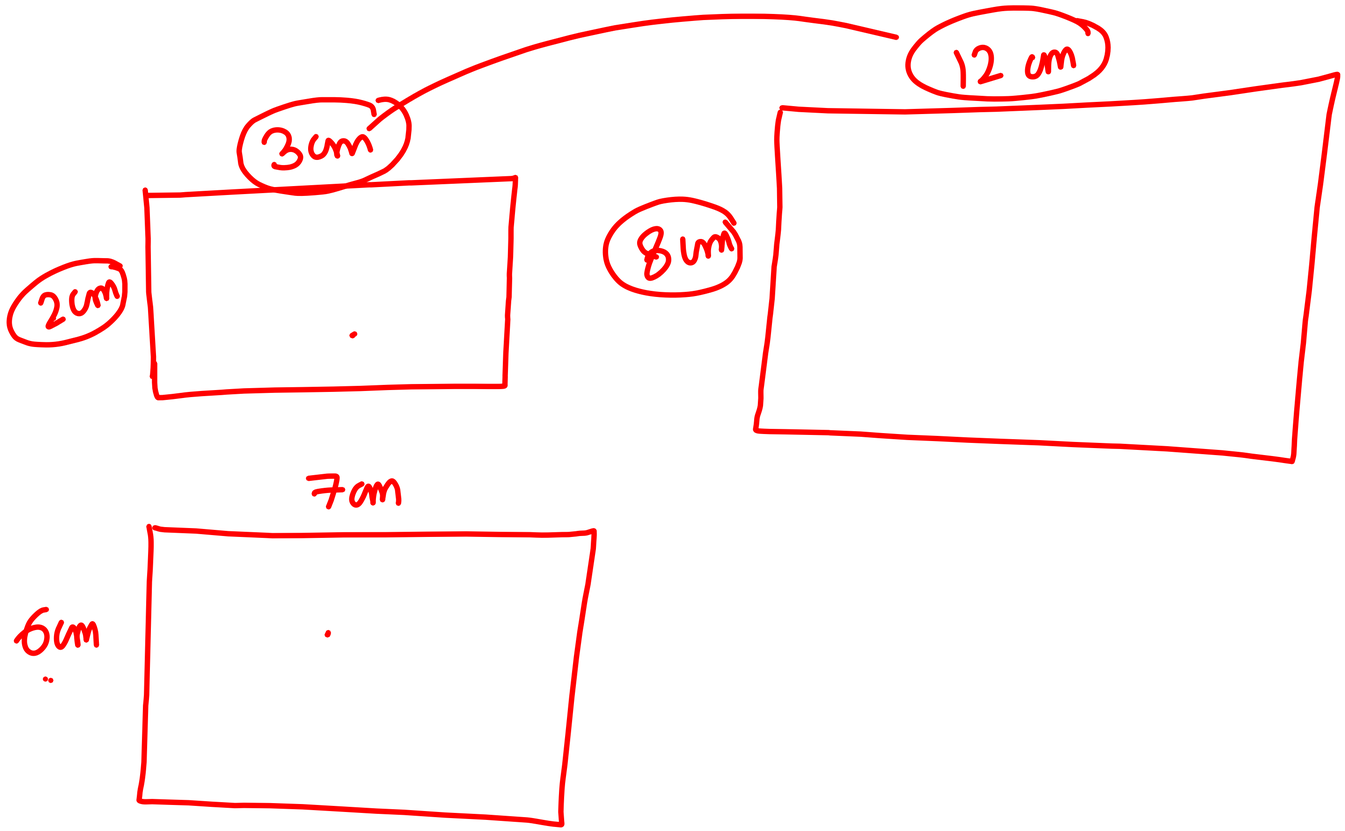
$\Delta ABC \sim \Delta PQR$ ✗

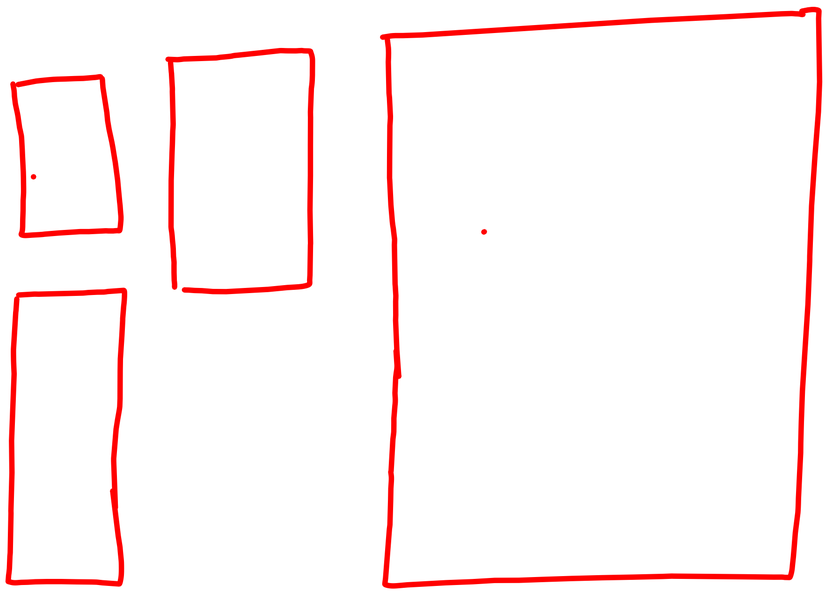
$\Delta ABC \sim \Delta QPR$ ✗





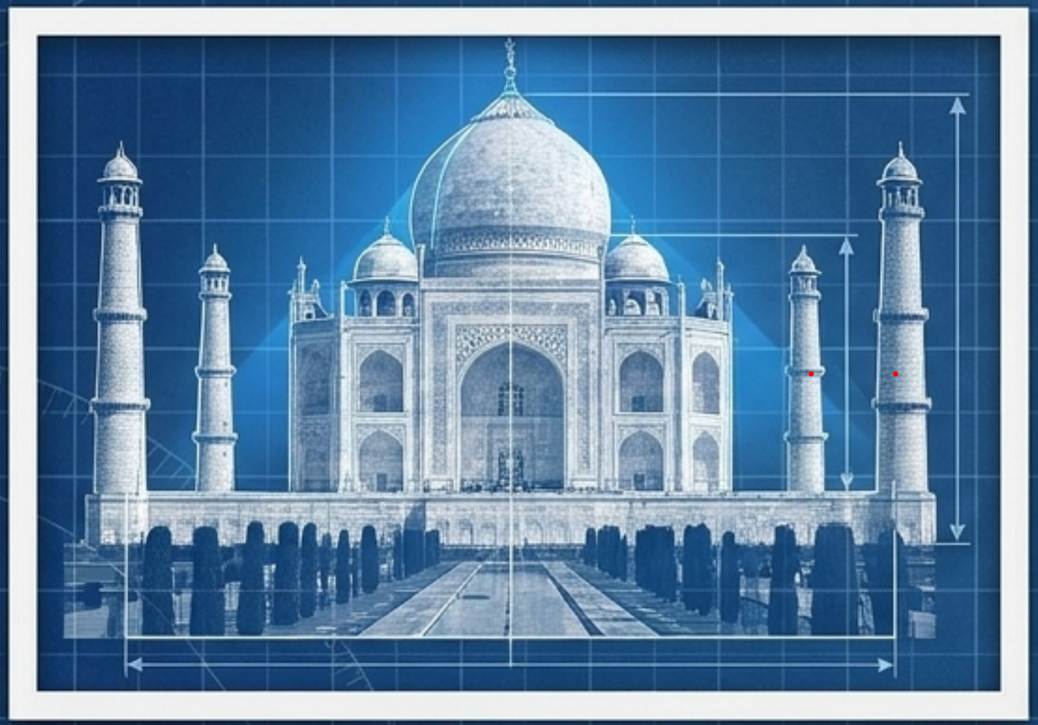
$\Delta ABC \sim \Delta QRP$


12 cm
3cm
2cm
8cm
7cm
6cm

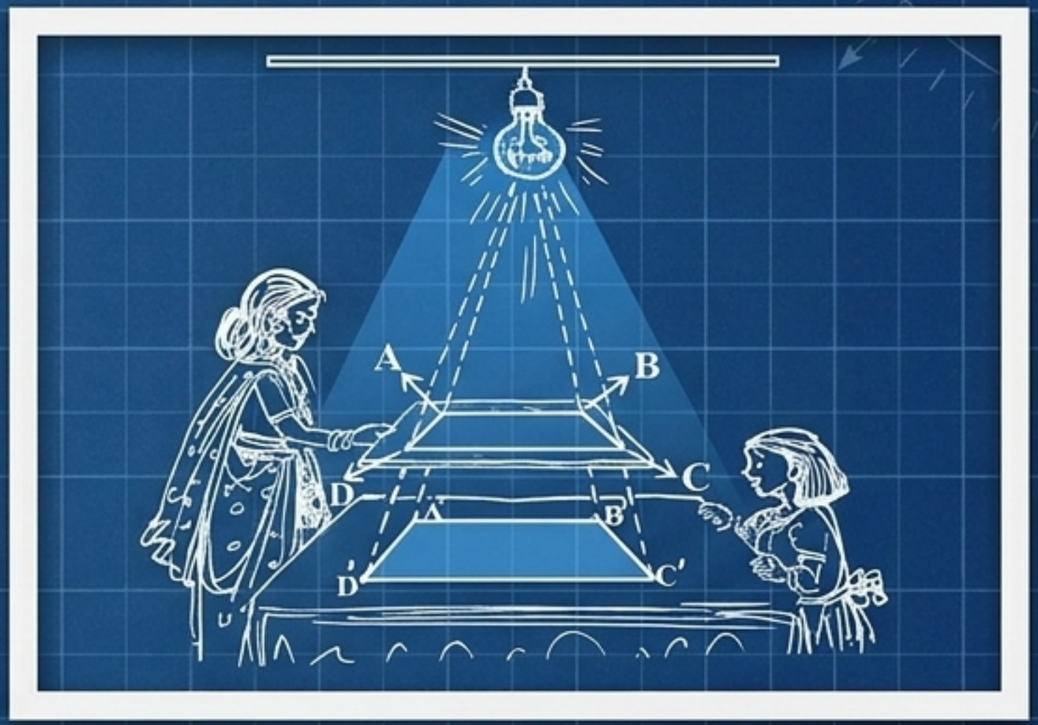


VISUALIZING SIMILARITY IN THE PHYSICAL WORLD

PERSPECTIVE

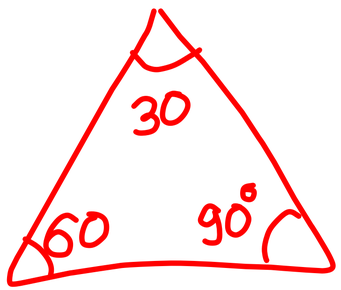


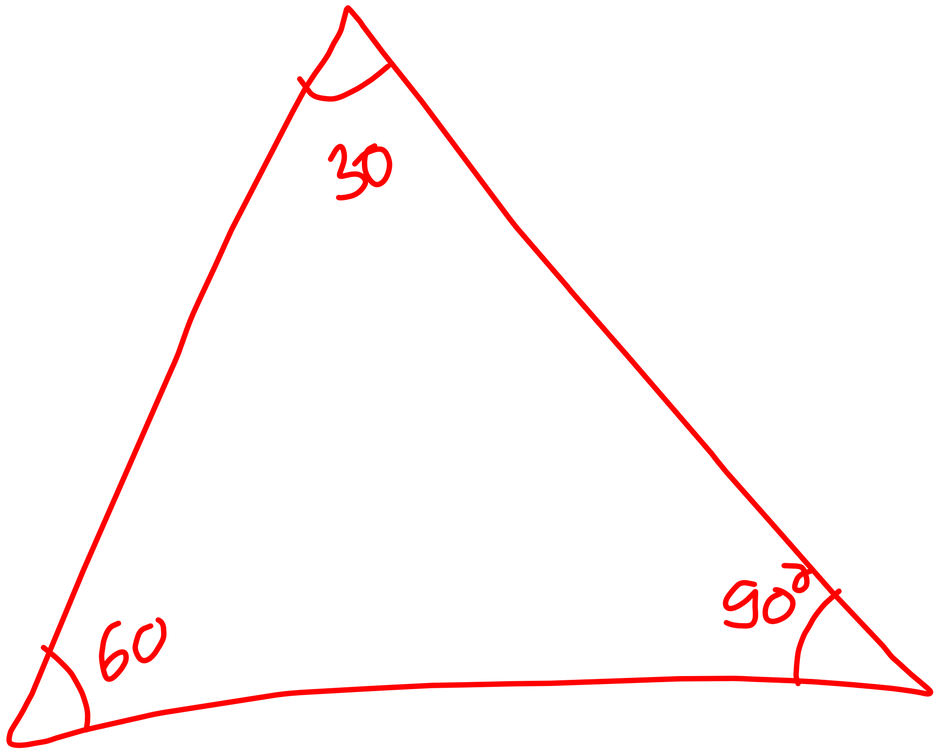
PROJECTION



An activity placing a lighted bulb above a cardboard quadrilateral casts a shadow on a table. The shadow A'B'C'D' is an enlargement (magnification) of the cardboard ABCD. The light propagates in a straight line, ensuring the shapes are similar. The Taj Mahal photograph demonstrates the same property: identical proportions at any scale factor.

Gemini Notebook


30
60
90°


30
60
90°

$\triangle ABC \sim \triangle PQR$

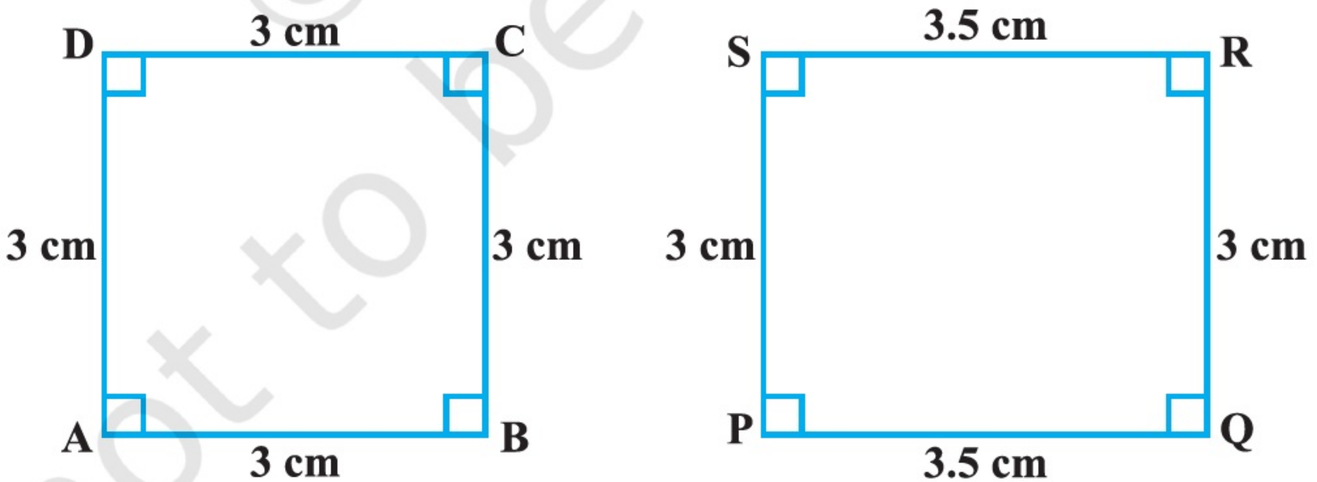
$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

By CPST

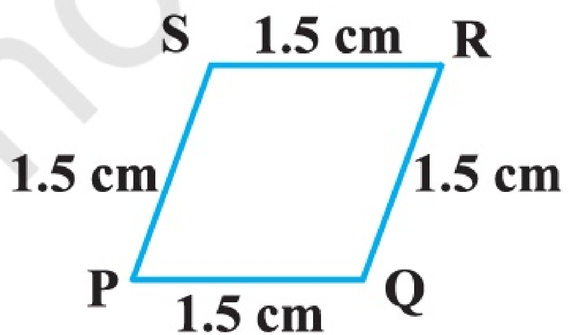
and

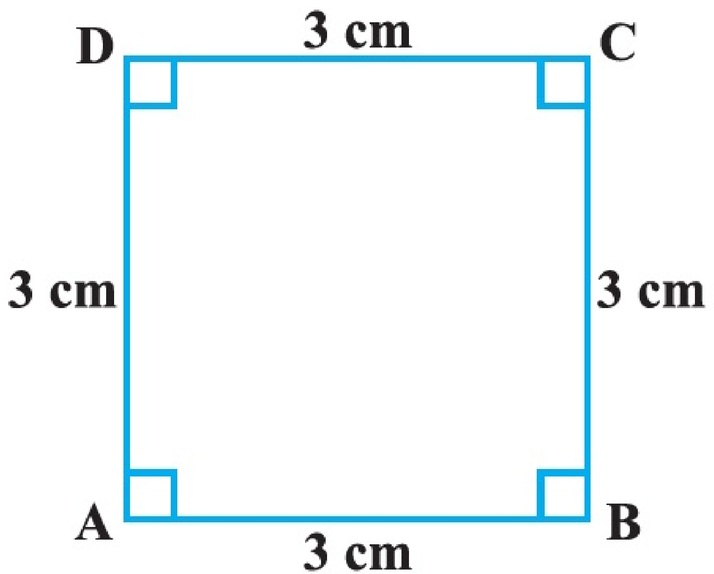
$\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$

State whether the following quadrilaterals are similar or not:

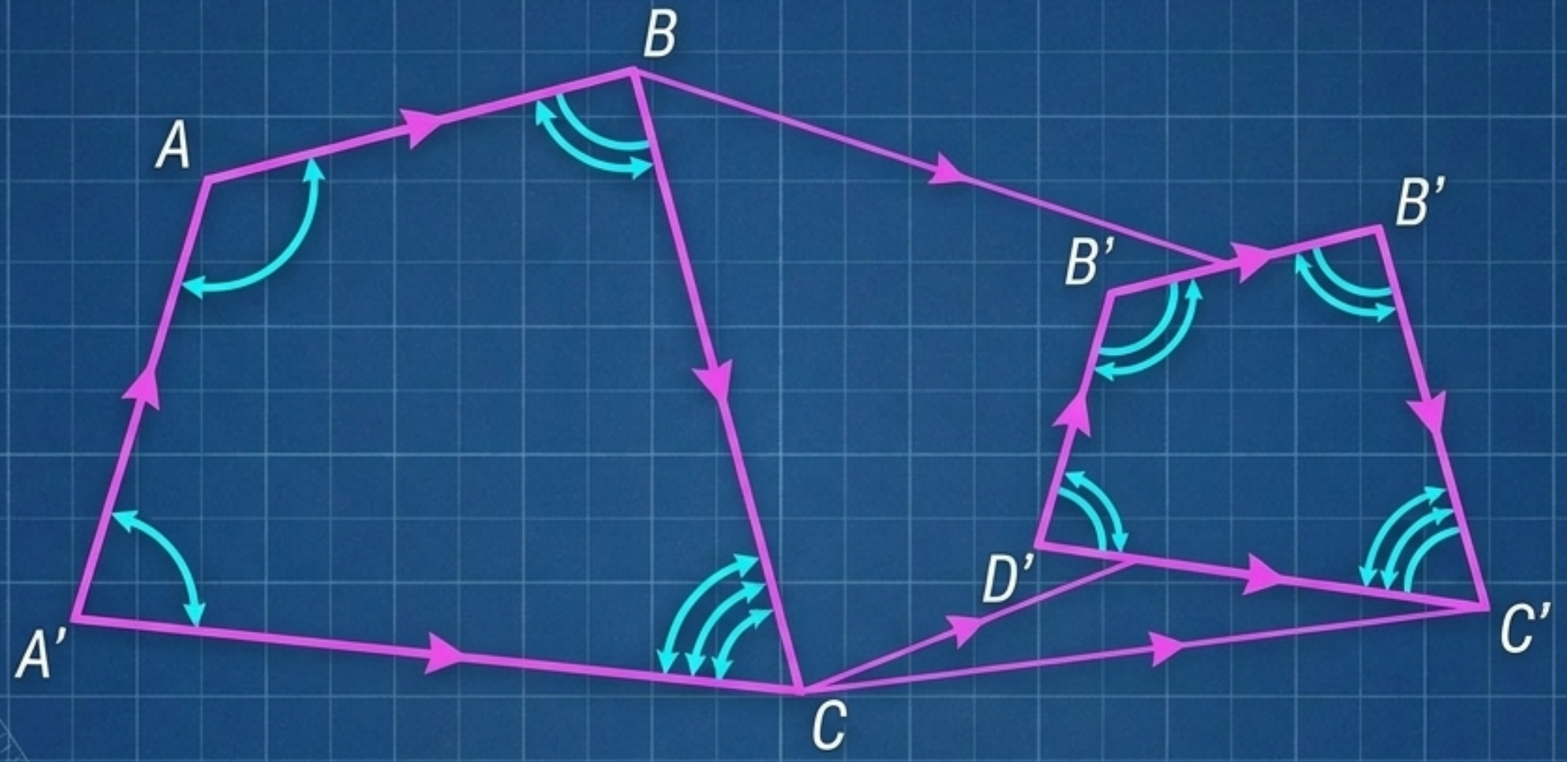


State whether the following quadrilaterals are similar or not:





THE TWO GOLDEN RULES OF POLYGON SIMILARITY



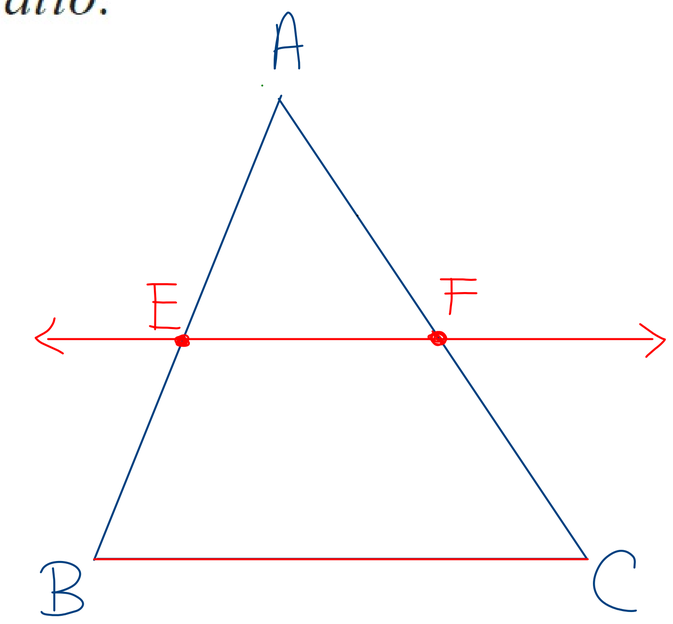
Two polygons of the same number of sides are similar if and only if two conditions are met:
1. Angle Condition: All their corresponding angles are equal.
2. Side Condition: All their corresponding sides are in the same ratio (proportional).

Note: The ratio of corresponding sides is referred to as the scale factor.

Gemini Notebook

BPT (Basic proportionality theorem)

Theorem 6.1 : *If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.*



If $EF \parallel BC$

then $\frac{AE}{EB} = \frac{AF}{FC}$

$$\frac{AE}{EB} = \frac{AF}{FC}$$

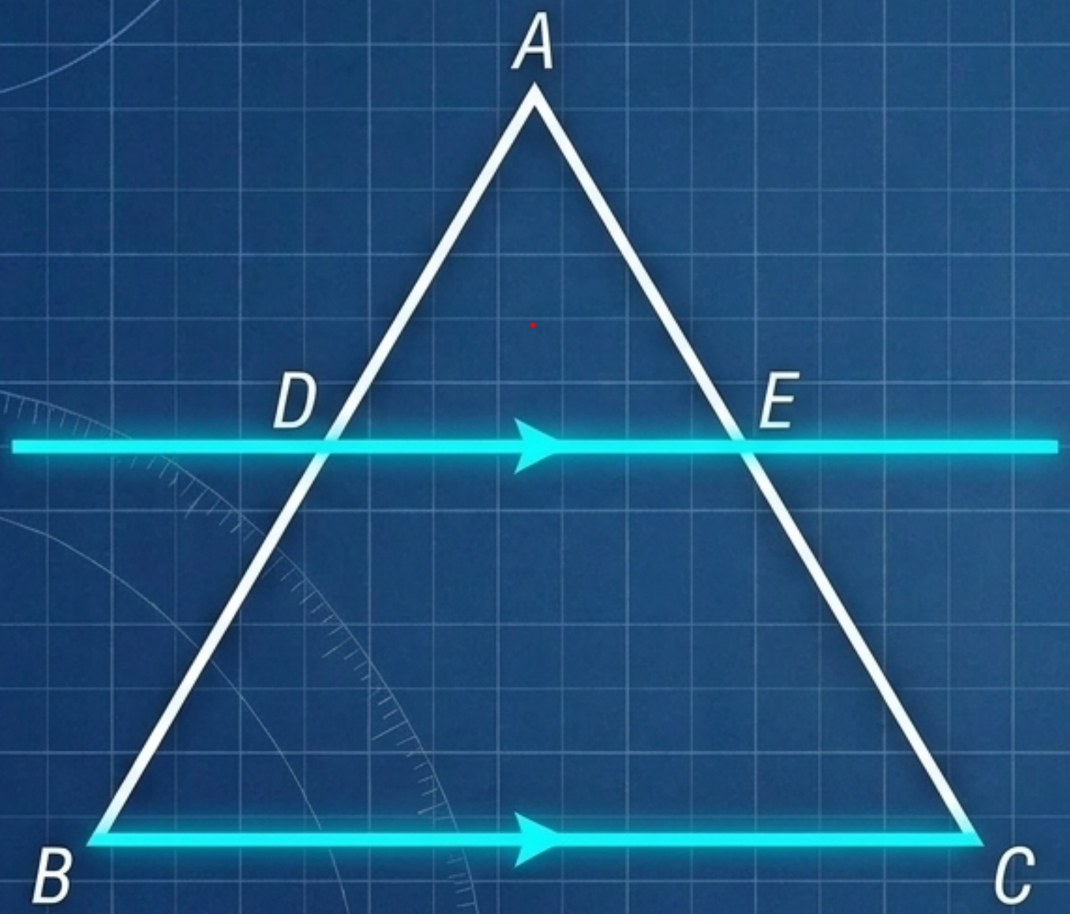
$$\frac{AE}{EB} + 1 = \frac{AF}{FC} + 1$$

$$\Rightarrow \frac{AE + EB}{EB} = \frac{AF + FC}{FC}$$

$$\Rightarrow \frac{AB}{EB} = \frac{AC}{FC}$$

$$\frac{AB}{AE} = \frac{AC}{AF}$$

BASIC PROPORTIONALITY THEOREM (THALES THEOREM)



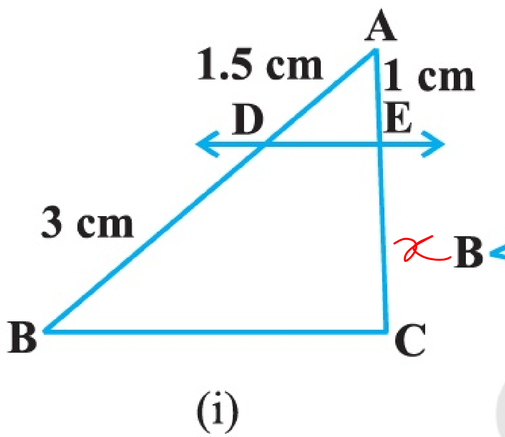
Theorem 6.1: If a straight line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.

The Setup: In triangle ABC, a straight line DE || BC, intersects AB at D and AC at E.

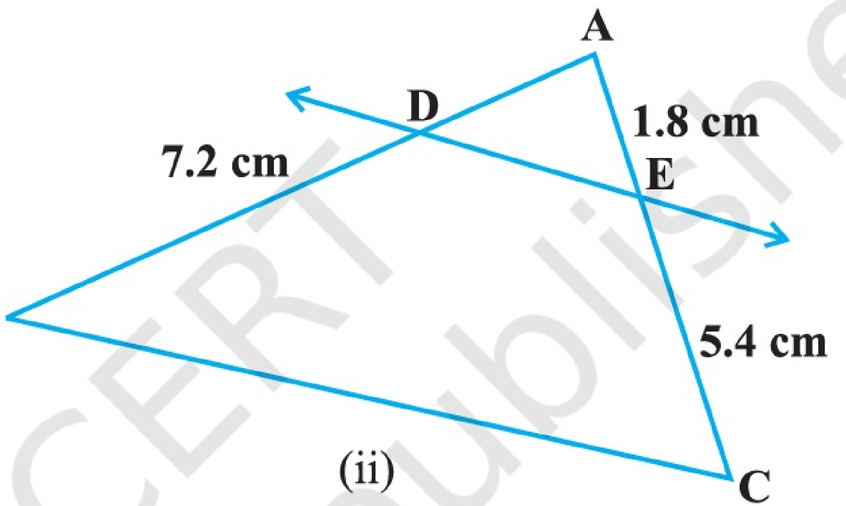
$$\frac{AB}{AD} = \frac{AC}{AE} \quad \text{(ii)} \quad \frac{AB}{DB} = \frac{AC}{EC}$$

Gemini Notebook

1. In Fig. 6.17, (i) and (ii), DE || BC. Find EC in (i) and AD in (ii).



(i)



(ii)

(i) $\frac{AD}{DB} = \frac{AE}{EC}$

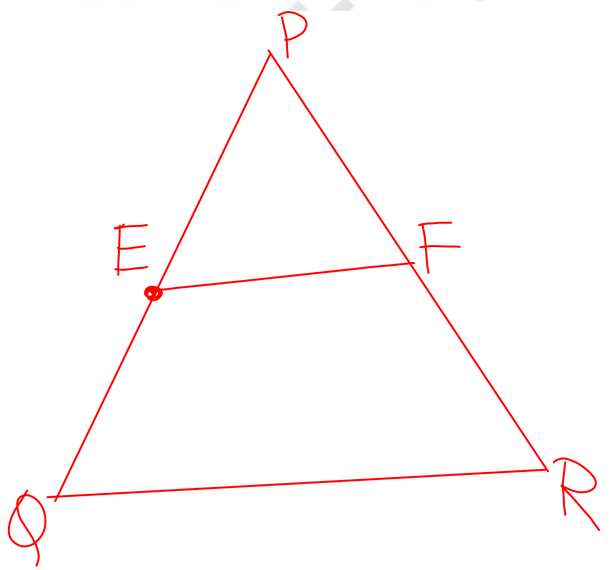
$\frac{1.5}{3} = \frac{1}{x}$

2. E and F are points on the sides PQ and PR respectively of a ΔPQR. For each of the following cases, state whether EF || QR :

(i) PE = 3.9 cm, EQ = 3 cm, PF = 3.6 cm and FR = 2.4 cm

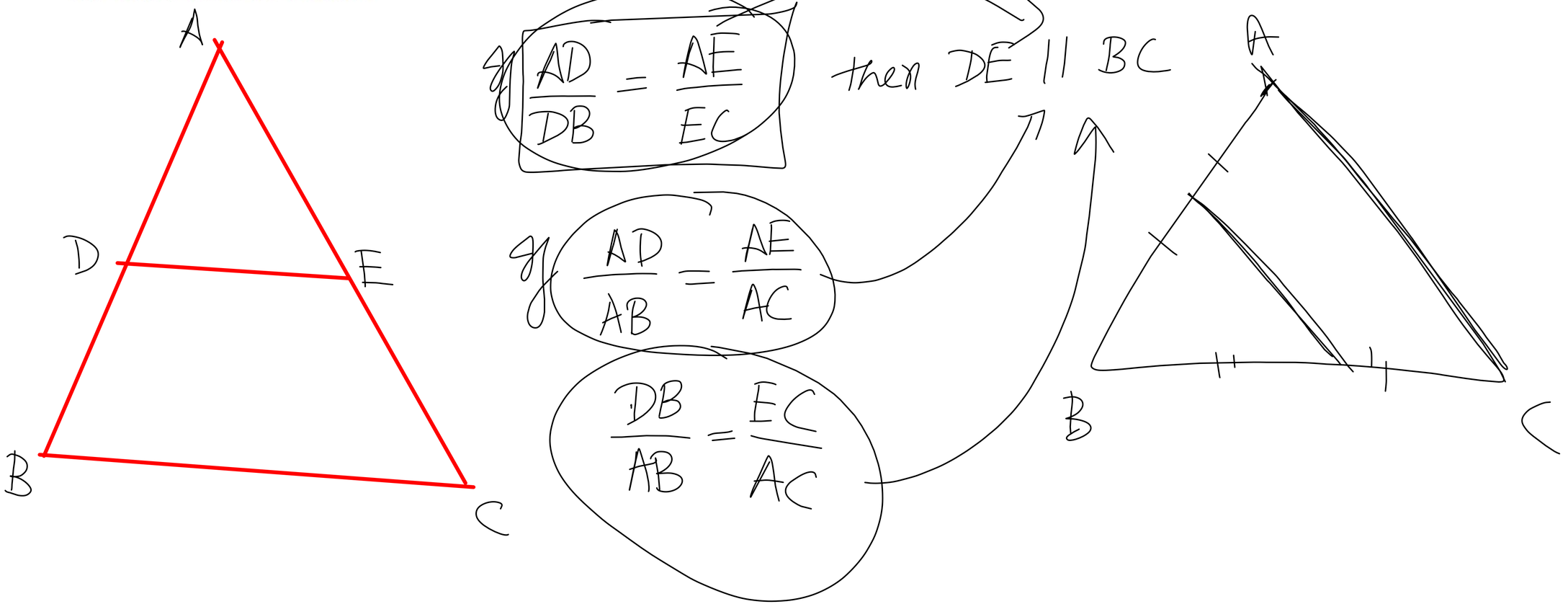
(ii) PE = 4 cm, QE = 4.5 cm, PF = 8 cm and RF = 9 cm

(iii) PQ = 1.28 cm, PR = 2.56 cm, PE = 0.18 cm and PF = 0.36 cm



Converse of BPT

Theorem 6.2 : *If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.*



3. In Fig. 6.18, if LM || CB and LN || CD, prove that

$$\frac{AM}{AB} = \frac{AN}{AD}.$$

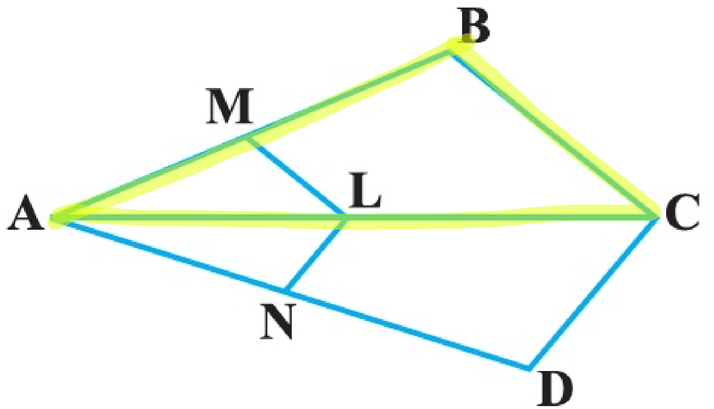


Fig. 6.18

In $\triangle ABC$

$ML \parallel BC$

then $\frac{AM}{AB} = \frac{AL}{AC}$ — (i)

In $\triangle ADC$

$LN \parallel CD$

then $\frac{AN}{AD} = \frac{AL}{AC}$ — (ii)

[By BPT]

from (i) & (ii) $\frac{AM}{AB} = \frac{AN}{AD}$

4. In Fig. 6.19, DE || AC and DF || AE. Prove that

$$\frac{BF}{FE} = \frac{BE}{EC}.$$

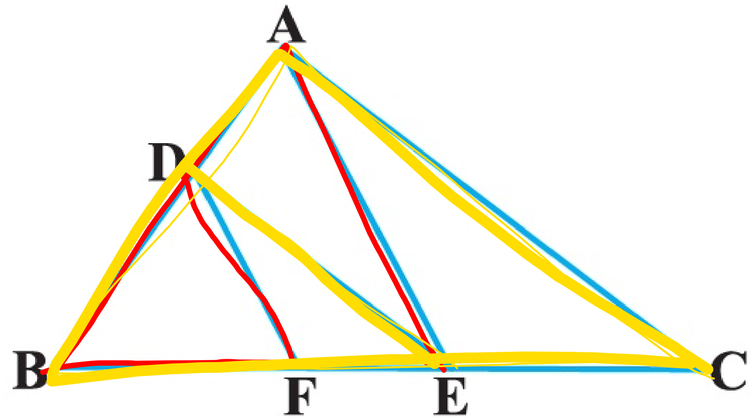


Fig. 6.19

if DE || AC

then $\frac{BD}{DA} = \frac{BE}{EC}$ (i)

By BPT

AE || DF

$\frac{BD}{DA} = \frac{BF}{FE}$ (By BPT) ... ii.

From (i) and (ii)

$\frac{BD}{DA} = \frac{BD}{DA}$

So, $\frac{BE}{EC} = \frac{BF}{FE}$

HP.

5. In Fig. 6.20, DE || OQ and DF || OR. Show that EF || QR.

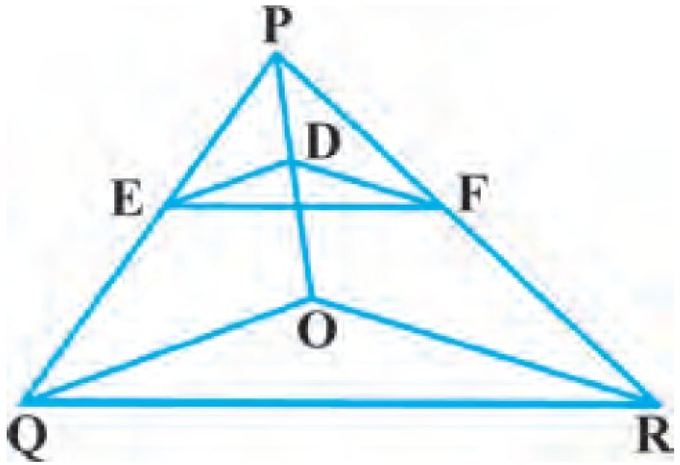


Fig. 6.20

Soln

In $\triangle PQO$

$ED \parallel QO$

So, $\frac{PE}{EQ} = \frac{PD}{DO}$ (?) —(i)

In $\triangle POR$

$DF \parallel OR$

So, $\frac{PD}{DO} = \frac{PF}{FR}$ (?) —(ii)

from (i) and (ii)

In $\triangle PQR$ → $\frac{PE}{EQ} = \frac{PF}{FR}$

By using converse of BPT $EF \parallel QR$.

6. In Fig. 6.21, A, B and C are points on OP, OQ and OR respectively such that AB || PQ and AC || PR. Show that BC || QR.

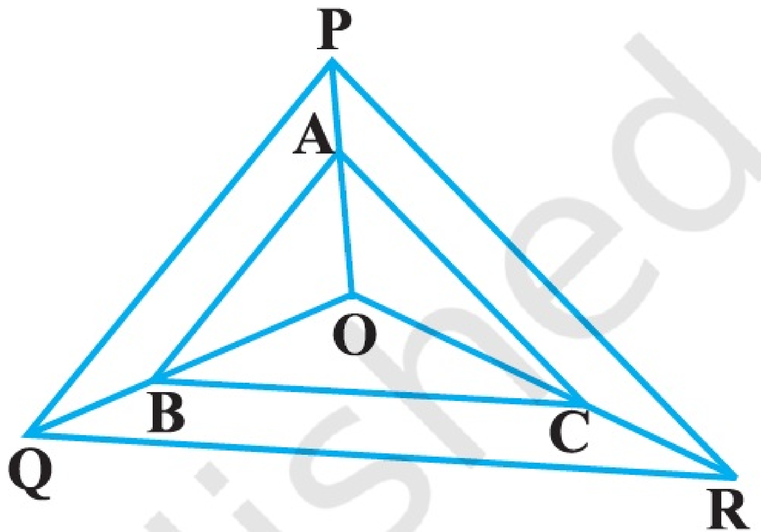


Fig. 6.21

In $\triangle POQ$

$AB \parallel PQ$.

So, $\frac{OA}{AP} = \frac{OB}{BQ}$ ...(i)

In $\triangle POR$

$AC \parallel PR$

So, $\frac{OA}{AP} = \frac{OC}{CR}$ ...(ii)

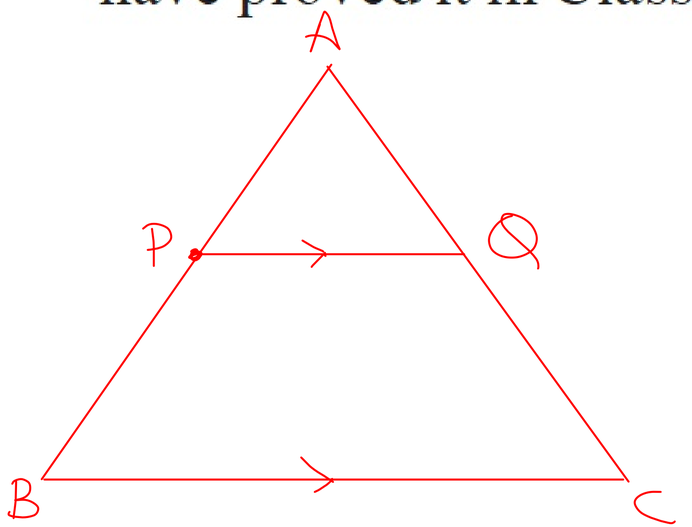
From (i) & (ii),

$\frac{OB}{BQ} = \frac{OC}{CR}$

$\therefore BC \parallel QR$ by converse of B.P.T

Published

7. Using Theorem 6.1, prove that a line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side. (Recall that you have proved it in Class IX).



Given

P is the mid-point of AB.

& PQ || BC

To prove

Q is the mid-point of AC

In $\triangle ABC$

$PQ \parallel BC$

So, $\frac{AP}{PB} = \frac{AQ}{QC}$

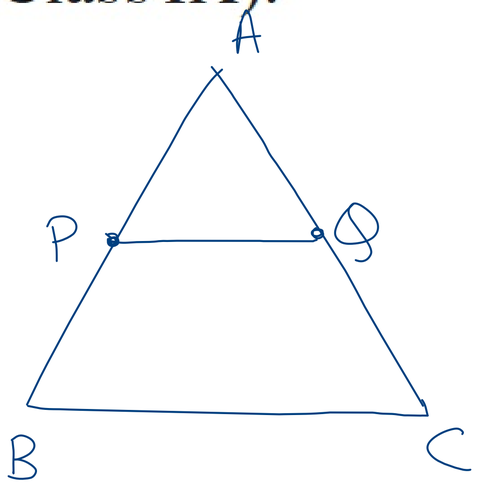
Since, P is the mid-point $\therefore AP = PB$.

$\frac{1}{1}\frac{\cancel{AP}}{\cancel{AP}} = \frac{AQ}{QC}$

$AQ = QC$ [Q is the mid-point of AC]

1) a_9

8. Using Theorem 6.2, prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side. (Recall that you have done it in Class IX).



Since P is the mid-point of AB so, $\frac{AP}{PB} = 1$ —(i)

Q

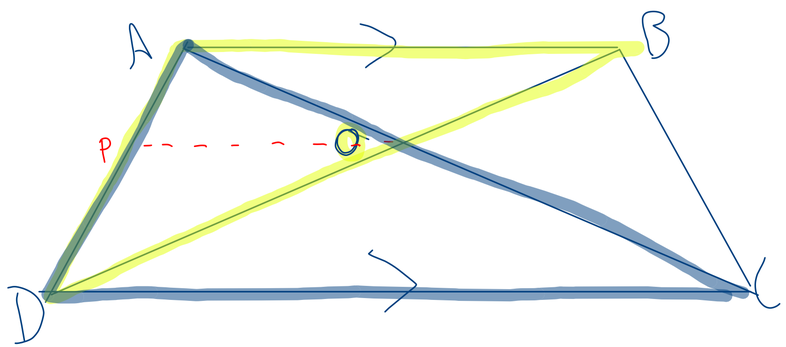
AC, so $\frac{AQ}{QC} = 1$ —(ii)

from (i) and (ii) we get $\frac{AP}{PB} = \frac{AQ}{QC}$

Now, By converse of BPT

$PQ \parallel BC$.

9. ABCD is a trapezium in which AB || DC and its diagonals intersect each other at the point O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$.



Construction of PO

PO || AB || DC

In $\triangle ABD$

$\Rightarrow$ PO || AB

$\Rightarrow$ By using BPT we can say

$\Rightarrow \boxed{\frac{DP}{PA} = \frac{DO}{BO}}$ (i)

In $\triangle DCA$

$\Rightarrow$ PO || DC

$\Rightarrow$ By using BPT we can say

$\Rightarrow \boxed{\frac{DP}{PA} = \frac{CO}{AO}}$ (ii)

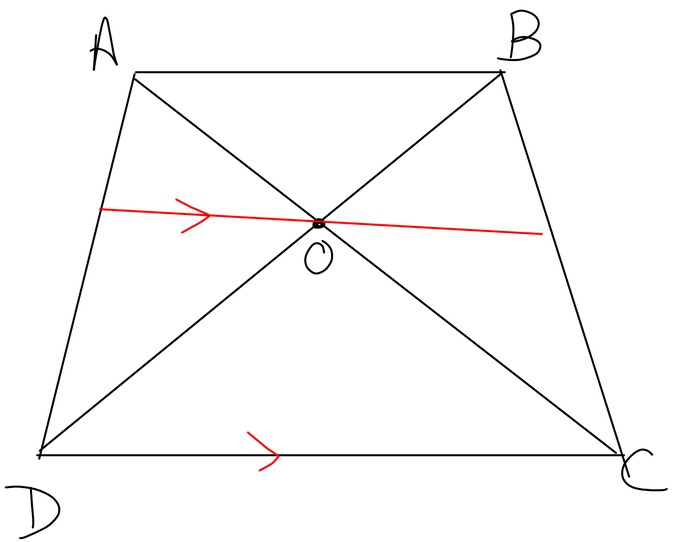
From (i) and (ii)

$\frac{CO}{AO} = \frac{DO}{BO}$

$\boxed{\frac{CO}{DO} = \frac{AO}{BO}}$

HP

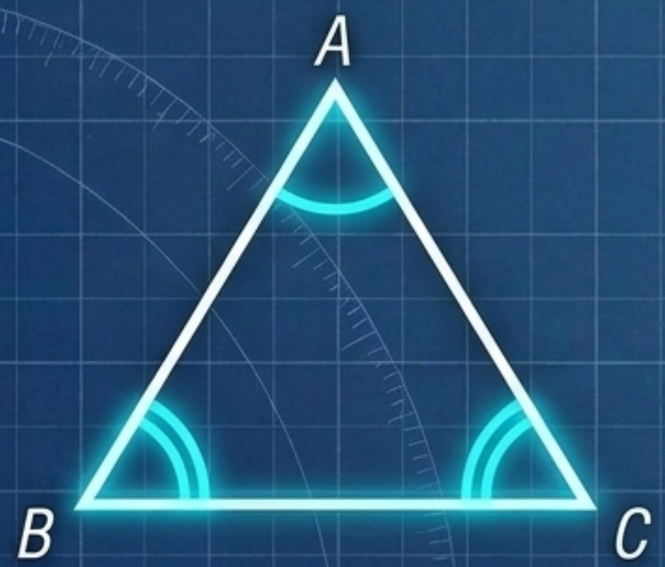
10. The diagonals of a quadrilateral ABCD intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that ABCD is a trapezium.



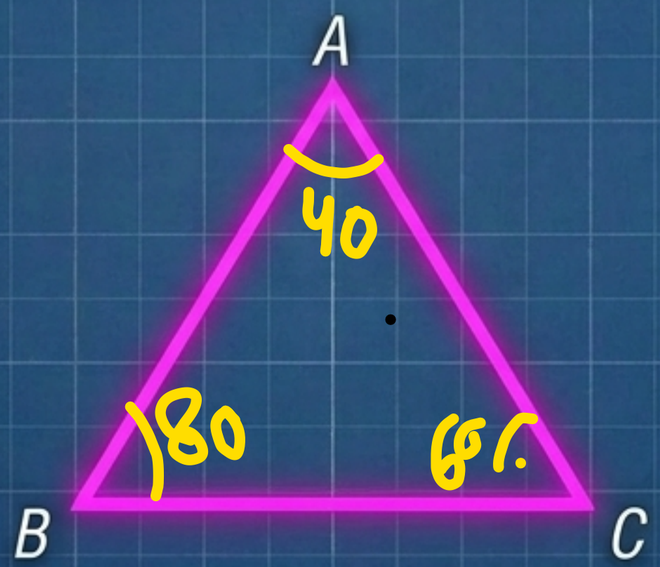
The Three Criteria for Triangle Similarity

For triangles, we do not need to check both polygon conditions independently. One condition mathematically implies the other. These three criteria are sufficient to prove similarity between triangle ABC and triangle DEF.

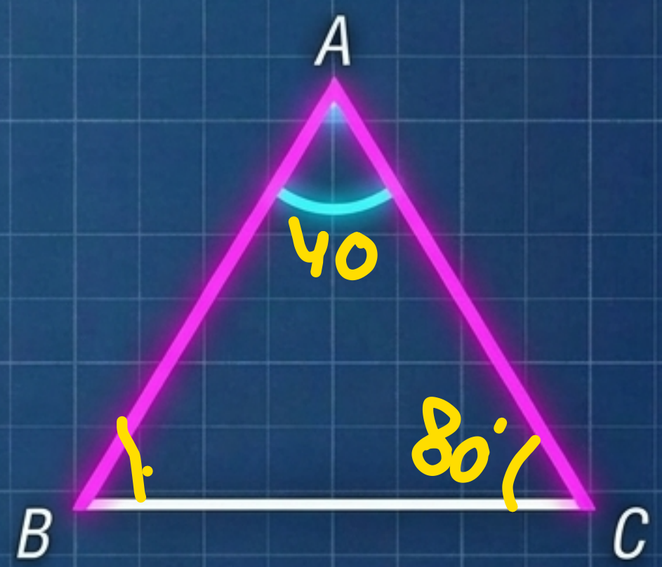
AAA / AA
(Angle-Angle-Angle)



SSS
(Side-Side-Side)

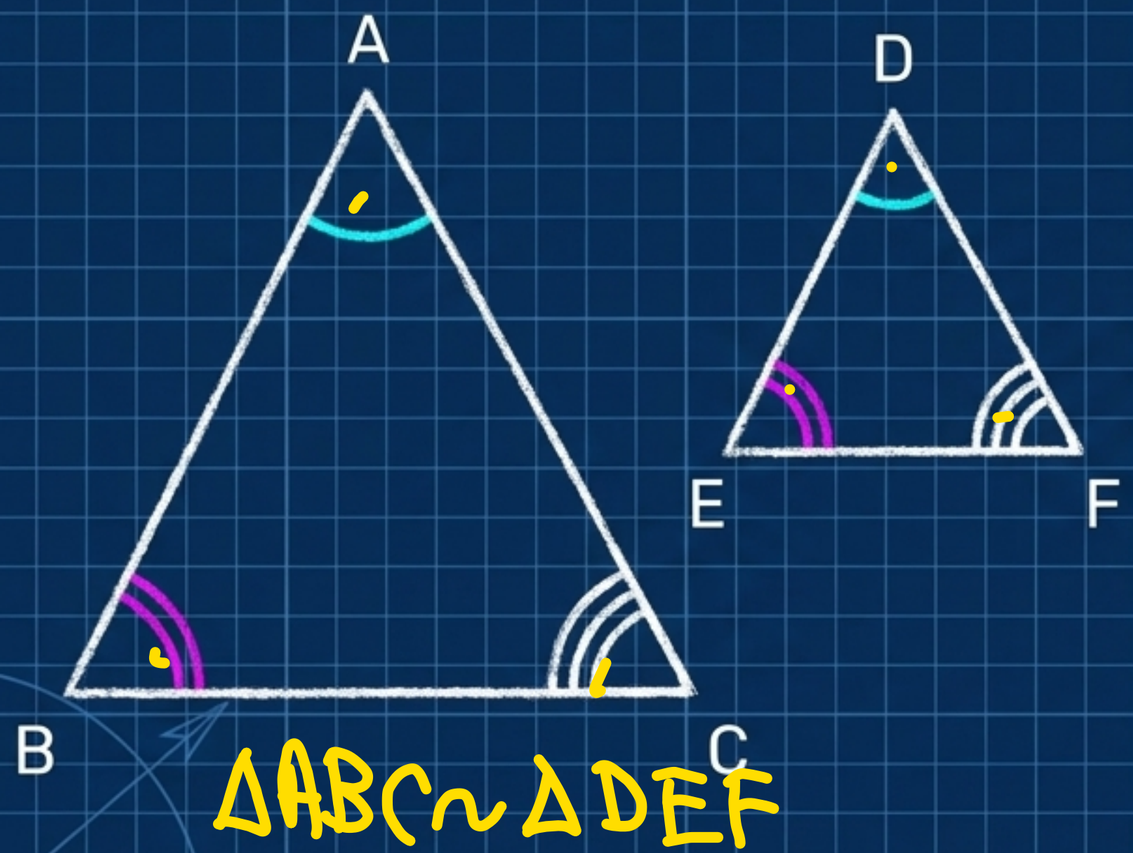


SAS
(Side-Angle-Side)



Gemini Notebook

The Angle Criteria: AAA and AA



AAA Criterion: If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio.

Result: Triangle ABC ~ Triangle DEF.

The AA Shortcut. If two angles of one triangle are respectively equal to two angles of another triangle, the third angles must be equal (due to the angle sum property). Thus, AA similarity is sufficient.

Gemini Notebook

Example 4 : In Fig. 6.29, if PQ || RS, prove that Δ POQ ~ Δ SOR.

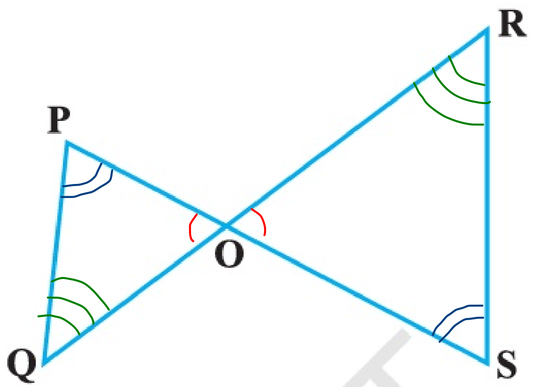


Fig. 6.29

In ΔPOQ and ΔROS

$\angle POQ = \angle ROS$ (V. opposite angles)

$\angle OPQ = \angle OSR$ {Alternate int. angles}

$\angle PQO = \angle ORS$

By AAA similarity criteria

$\Delta POQ \sim \Delta SOR$

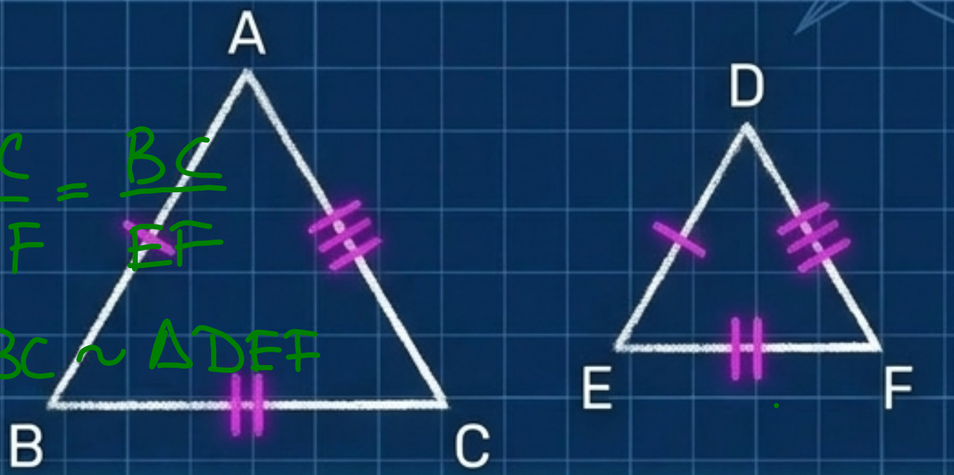
The Side Criteria: SSS and SAS

SSS Criterion

If the sides of one triangle are proportional to the sides of another triangle, then their corresponding angles are equal. Hence, the two triangles are similar.

if $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$

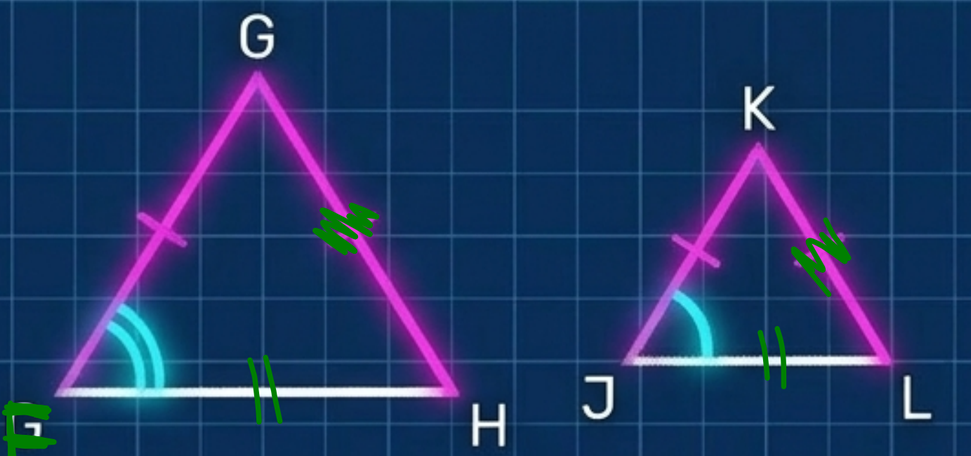
then $\triangle ABC \sim \triangle DEF$



SAS Criterion

If one angle of a triangle is equal to one angle of the other triangle, and the corresponding sides including these angles are proportional, then the triangles are similar.

if $\frac{FG}{JK} = \frac{FH}{JL}$ and $\angle F = \angle J$



Gemini Notebook

EXERCISE 6.3

1. State which pairs of triangles in Fig. 6.34 are similar. Write the similarity criterion used by you for answering the question and also write the pairs of similar triangles in the symbolic form :

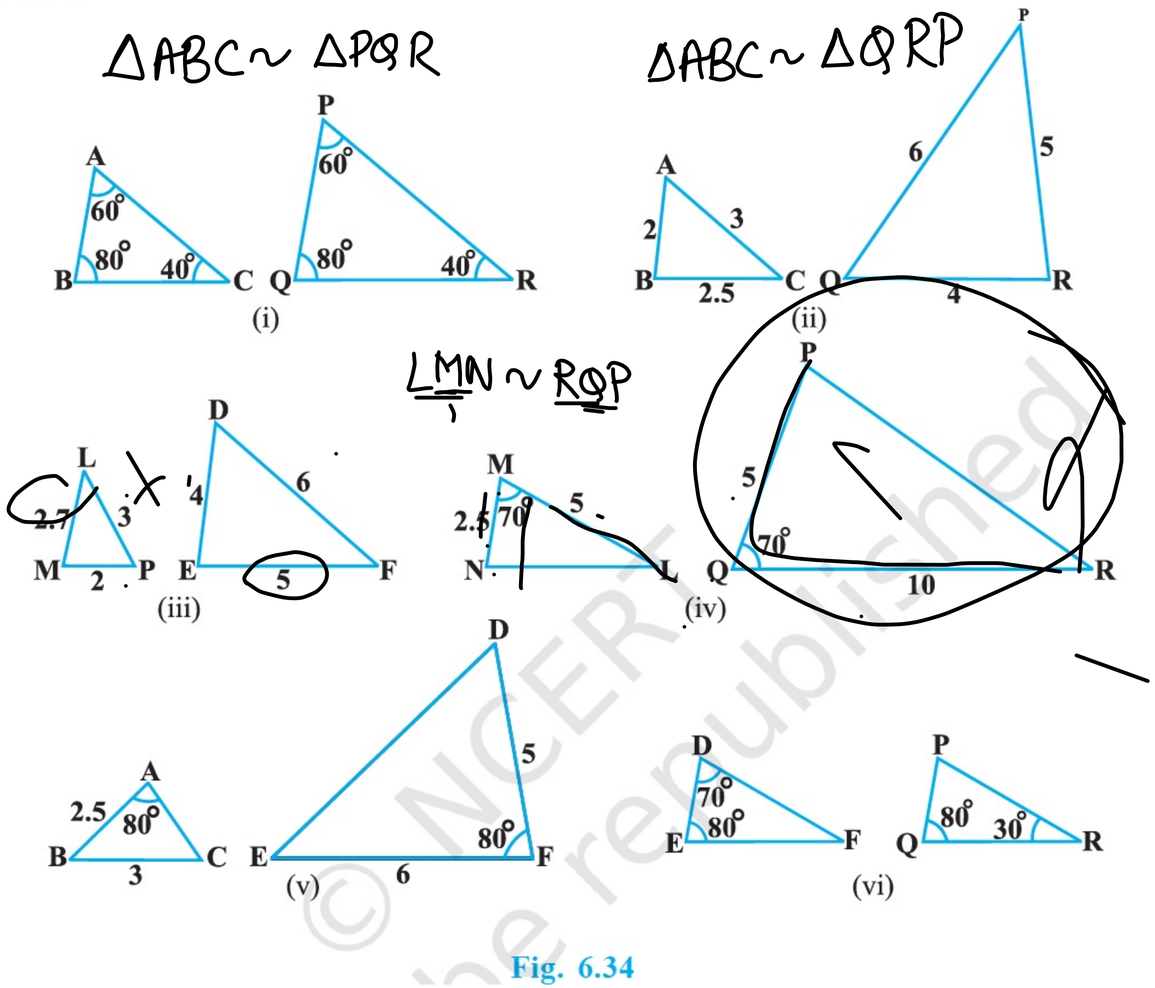


Fig. 6.34

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Example 5 : Observe Fig. 6.30 and then find ∠ P.

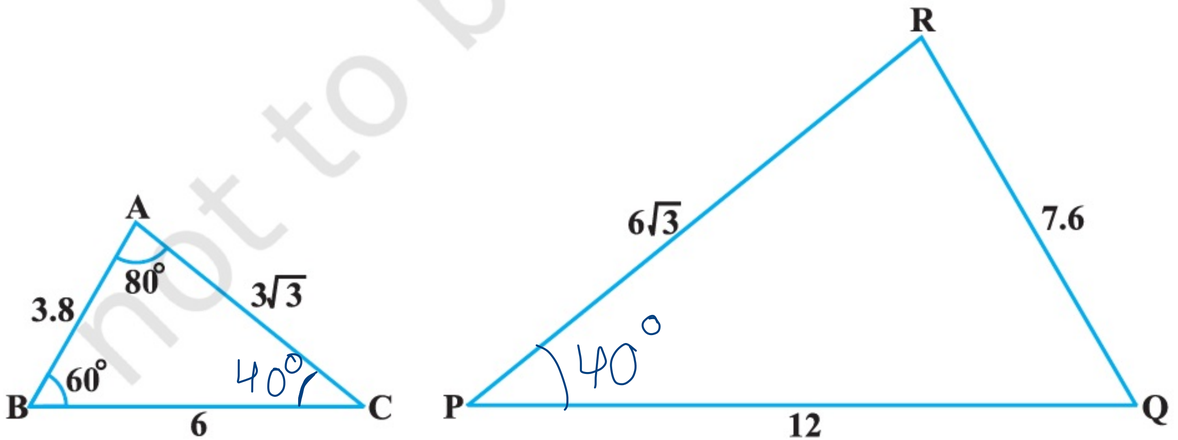


Fig. 6.30

By SSS similarity criteria

$\Delta ABC \sim \Delta RQP$

$\frac{AB}{RQ} = \frac{BC}{PQ} = \frac{AC}{RP}$

By CPST $\angle P = \angle C$

2. In Fig. 6.35, $\Delta ODC \sim \Delta OBA$, $\angle BOC = 125°$ and $\angle CDO = 70°$. Find $\angle DOC$, $\angle DCO$ and $\angle OAB$.

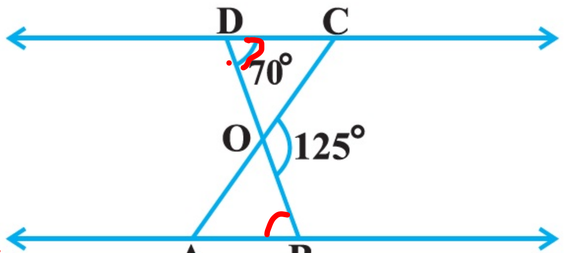


Fig. 6.35

$\Delta ODC \sim \Delta OBA$

By CPST

$\angle DCO = \angle BAO$

$= 55°$

$\angle COB + \angle DOC$ [Linear Pair]

$125 + \angle DOC = 180°$

$\angle DOC = 180 - 125 = 55°$

By using Angle Sum Prop

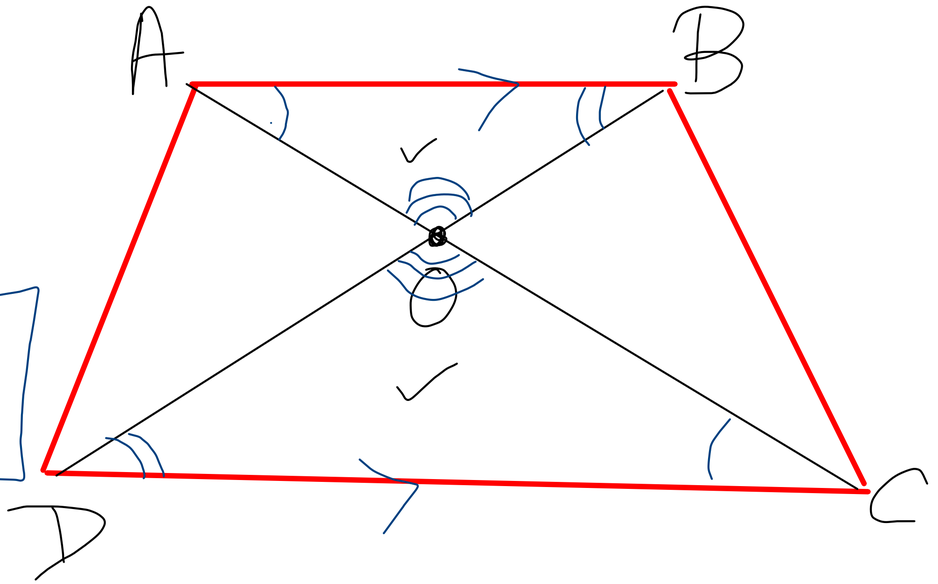
$\angle DOC + \angle ODC + \angle OCD = 180°$

$55° + 70° + \angle OCD = 180°$

$\angle OCD = 180 - 125$

$\angle OCD = 55$

3. Diagonals AC and BD of a trapezium ABCD with AB || DC intersect each other at the point O. Using a similarity criterion for two triangles, show that $\frac{OA}{OC} = \frac{OB}{OD}$.



$\frac{OA}{OB} = \frac{OC}{OD}$

In $\triangle AOB$ and $\triangle DOC$

$\angle OAB = \angle OCD$

$\angle OBA = \angle ODC$

By AA similarity criteria

$\triangle AOB \sim \triangle COD$

By CPST

$\frac{OA}{OC} = \frac{OB}{OD}$

4. In Fig. 6.36, $\frac{QR}{QS} = \frac{QT}{PR}$ and $\angle 1 = \angle 2$. Show that $\Delta PQS \sim \Delta TQR$.

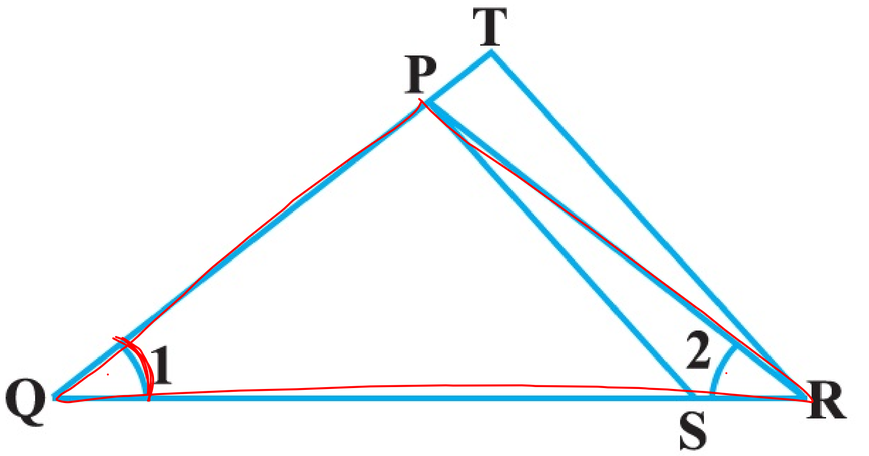


Fig. 6.36

ΔPQR is an isoceles triangle ($\angle 1 = \angle 2$)

Now, $PQ = PR$

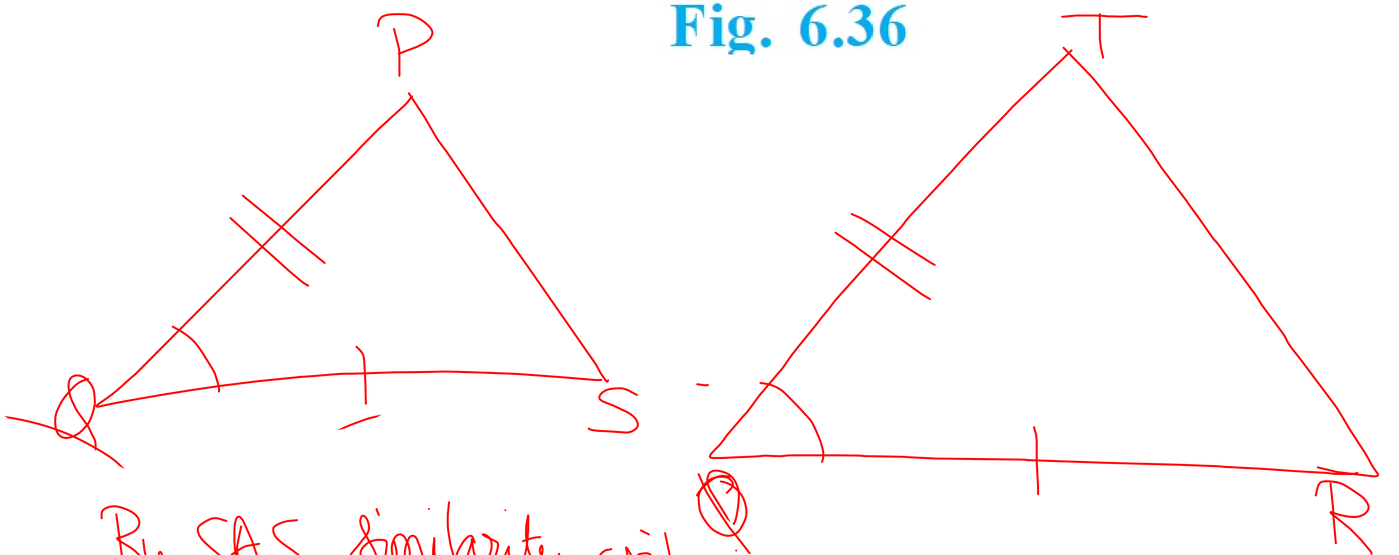
Given $\frac{QR}{QS} = \frac{QT}{PR}$

↓

Replace PR with PQ

$\frac{QR}{QS} = \frac{QT}{PQ}$

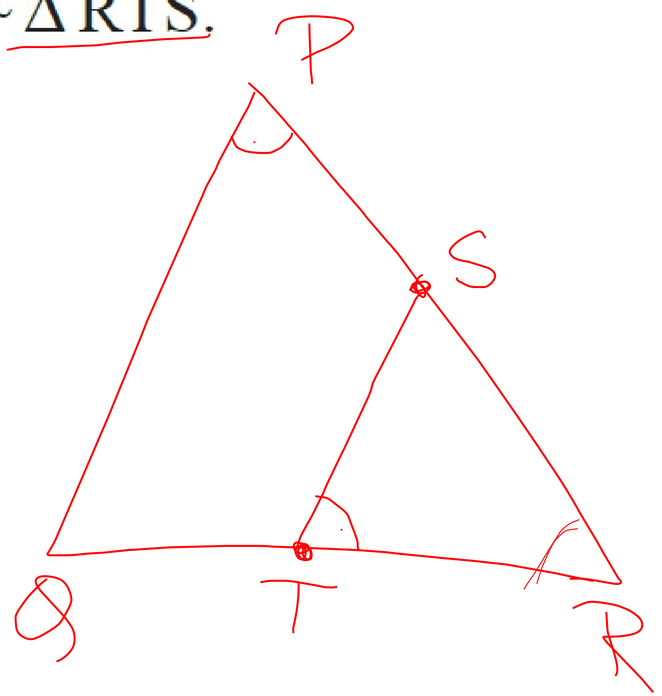
$\angle PQS = \angle TQR$ [common]



By SAS similarity criteria

$\Delta PQS \sim \Delta TQR$

5. S and T are points on sides PR and QR of Δ PQR such that $\angle$ P = $\angle$ RTS. Show that Δ RPQ ~ Δ RTS.



6. In Fig. 6.37, if Δ ABE $\cong$ Δ ACD, show that Δ ADE ~ Δ ABC.

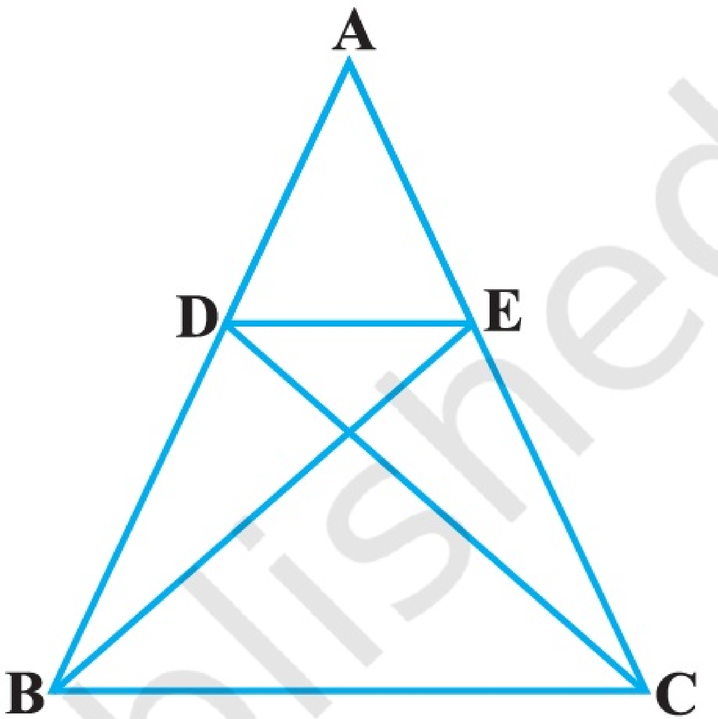


Fig. 6.37

7. In Fig. 6.38, altitudes AD and CE of ΔABC intersect each other at the point P. Show that:

(i) $\Delta AEP \sim \Delta CDP$

(ii) $\Delta ABD \sim \Delta CBE$

(iii) $\Delta AEP \sim \Delta ADB$

(iv) $\Delta PDC \sim \Delta BEC$

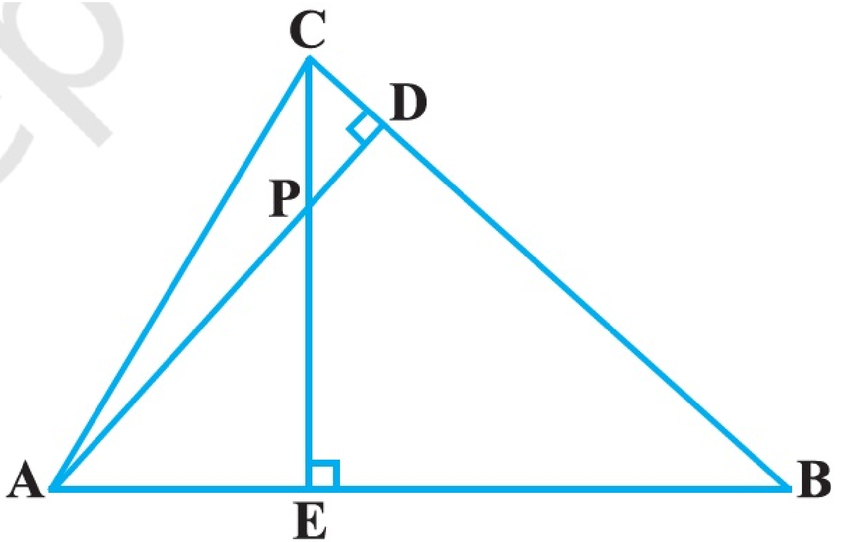


Fig. 6.38

8. E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that Δ ABE ~ Δ CFB.

9. In Fig. 6.39, ABC and AMP are two right triangles, right angled at B and M respectively. Prove that:

(i) $\Delta ABC \sim \Delta AMP$

(ii) $\frac{CA}{PA} = \frac{BC}{MP}$

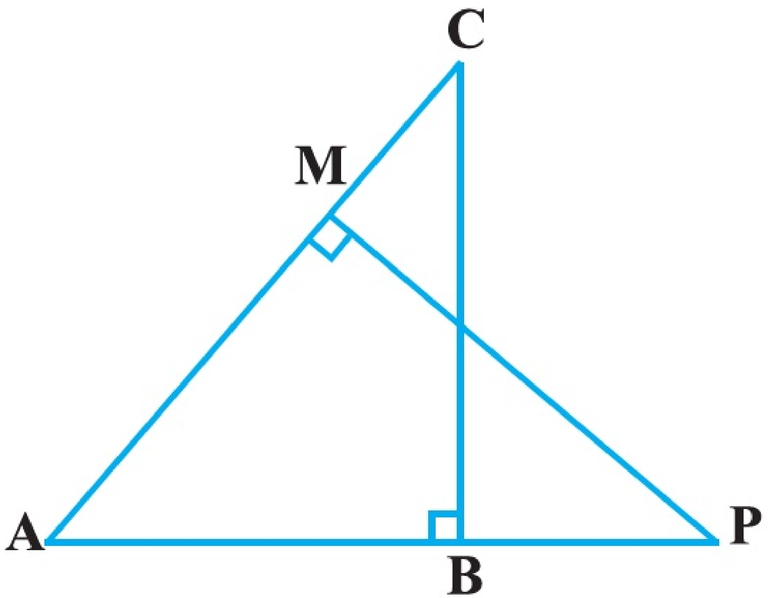


Fig. 6.39

10. CD and GH are respectively the bisectors of $\angle ACB$ and $\angle EGF$ such that D and H lie on sides AB and FE of ΔABC and ΔEFG respectively. If $\Delta ABC \sim \Delta FEG$, show that:

(i) $\frac{CD}{GH} = \frac{AC}{FG}$

(ii) $\Delta DCB \sim \Delta HGE$

(iii) $\Delta DCA \sim \Delta HGF$

11. In Fig. 6.40, E is a point on side CB produced of an isosceles triangle ABC with AB = AC. If AD ⊥ BC and EF ⊥ AC, prove that Δ ABD ~ Δ ECF.

12. Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of Δ PQR (see Fig. 6.41). Show that Δ ABC ~ Δ PQR.

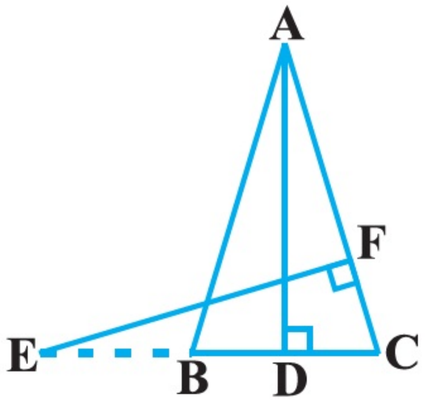


Fig. 6.40

13. D is a point on the side BC of a triangle ABC such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB.CD$.

14. Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that $\Delta ABC \sim \Delta PQR$.

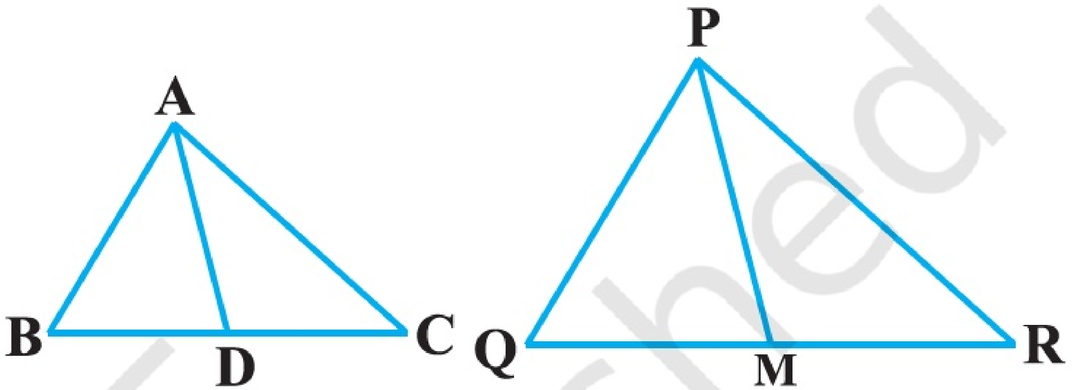


Fig. 6.41

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15. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

16. If AD and PM are medians of triangles ABC and PQR, respectively where $\Delta ABC \sim \Delta PQR$, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$.